Optical pH Sensor for Hydroponic Systems

Team Agro

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Bachelor of Science Honours in Information Technology and Management

Microcontroller-Based ICT Project (IS1901)

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1. Introduction

Measuring pH is an essential process in hydroponics and precision agriculture, as it directly determines the availability of dissolved nutrients to plants. Without accurate and continuous pH monitoring, hydroponic nutrient solutions can drift outside the optimal growing range of pH 5.5 to 6.5, causing nutrient lockout, stunted growth, and crop loss.

The primary challenge in the current landscape is that commercial electrochemical pH probes are expensive, fragile, and have a notoriously short operational lifetime due to glass electrode degradation and reference junction fouling in nutrient solutions. This project details the design and implementation of a custom-built, microcontroller-based optical pH sensor. By using colorimetric detection with a mixed Bromothymol Blue and Methyl Red indicator, a TCS34725 RGB sensor, and an ESP32 controller, this project creates a durable, cost-effective, and repairable alternative to commercial pH electrodes.

2. Aim and Objectives

Aim: To design, develop, and validate a low-cost optical pH sensor for continuous monitoring of hydroponic nutrient solutions using dual LED colorimetry, an ESP32 microcontroller, and DIY chemical indicators, achieving target measurement accuracy of ±0.025–0.03 pH at a cost reduction of over 90% compared to commercial electrochemical pH sensors.

Objectives:

- Design and fabricate a modular 5-component flow cell assembly (55×32×28mm) comprising four PETG 3D-printed parts and two separate acrylic optical windows, featuring dual LED illumination and a 30mm open optical path.
- Implement dual LED configuration using warm white (3000K) and cool white (6500K) LEDs to provide broad-spectrum illumination for both Bromothymol Blue and Methyl Red indicators simultaneously.
- Develop an ESP32-based control system with an automated Finite State Machine (FSM) controlling sample handling through three peristaltic pumps covering fill, dose, measure, drain, and rinse cycles.

- Integrate a TCS34725 RGB color sensor (MD0670) for high-resolution colorimetric detection with baseline correction and piecewise calibration against three pH buffer solutions.
- Formulate and prepare a DIY pH indicator solution combining Bromothymol Blue (0.1% stock) and Methyl Red (0.1% stock) with preservatives, achieving a shelf life of 6–9 months at room temperature.
- Calibrate the sensor system using standard pH buffer solutions (pH 4.01 and 6.86, with optional pH 9.18 for extended characterization) and validate accuracy using real-time temperature compensation via a DS18B20 sensor.
- Achieve target specifications: ±0.025–0.03 pH accuracy (design target, pending 100-cycle validation), operational pH 4.0–7.6 measurement range, and less than 30 seconds per measurement cycle.

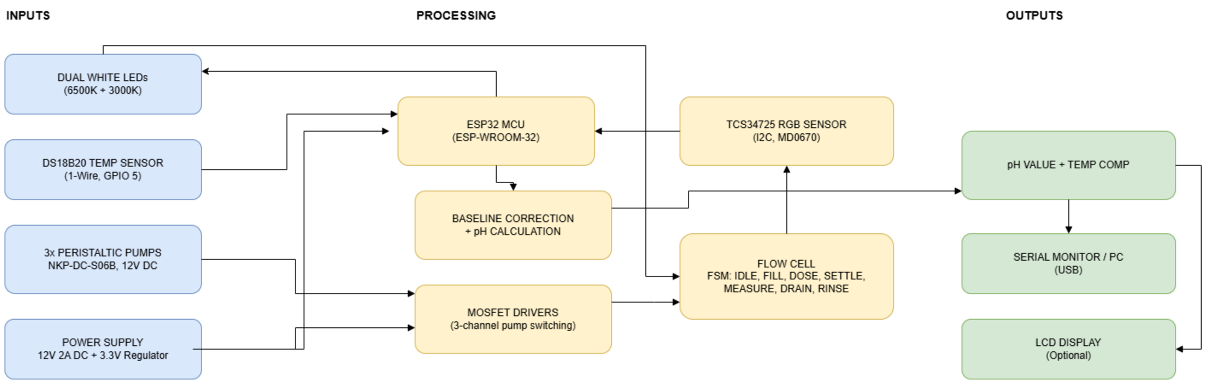
3. Problem and Proposed Solution

Problem: Standard commercial pH electrodes available on the market have a notoriously short lifetime, typically lasting 6–18 months before the glass bulb deteriorates, the reference junction becomes clogged, or calibration drift renders readings unreliable. In nutrient-rich hydroponic solutions, this degradation is accelerated by organic contamination and ionic interference, making continuous electrode-based pH monitoring a costly and maintenance-heavy investment for small-scale farmers.

Proposed Solution: The proposed solution is a custom optical pH sensor based on transmissive colorimetry. A mixed Bromothymol Blue and Methyl Red indicator solution is injected into a custom 3D-printed flow cell, and a calibrated dual white LED system illuminates the coloured sample. A TCS34725 RGB sensor reads the transmitted light and an ESP32 calculates the pH from the resulting color shift using the Beer–Lambert law. Because there is no glass electrode or reference junction, the sensor has no degrading electrochemical components. For 10.25 mL chamber operation, indicator consumption is approximately Rs. 0.23 per test compared to Rs. 15 per test for commercial dye, and the entire flow cell can be opened and cleaned in under five minutes.

4. Implementation

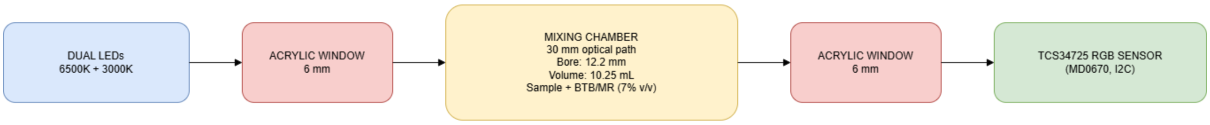
Complete Block Diagram



Supportive Internal Diagrams

The core of the hardware implementation relies on a transmissive optical measurement cell. Because the TCS34725 RGB sensor cannot measure pH directly, a pH-sensitive chemical indicator (Bromothymol Blue + Methyl Red) is mixed with the water sample inside the flow cell. The dual white LEDs illuminate the colored sample, and the RGB sensor reads the transmitted light intensity. The change in color across the red, green, and blue channels is used to calculate pH using the Color Index (CI) metric and piecewise linear calibration.

Optical Path Diagram



Light travels left to right through the colored liquid sample (Beer-Lambert absorbance).

How it Works (Calculations)

1. Color Index (CI): The ESP32 reads **`16-bit`** raw R, G, B values from the TCS34725 after subtracting the plain-water baseline (**`B_R`**, **`B_G`**, **`B_B`**). The Color Index represents color shift from red (acidic) toward green/blue (less acidic).

$$\text{CI} = (\text{G\_corrected} - \text{R\_corrected}) / (\text{G\_corrected} + \text{R\_corrected} + \text{B\_corrected})$$

Brief explanation: CI is normalized, so brightness changes affect the result less than using raw RGB values directly.

2. Temperature-Corrected Buffer References: Buffer pH values shift slightly with temperature. The DS18B20 measures real-time temperature (**T**°C), and firmware updates the reference pH values before interpolation.

$$\text{pH}_4\text{\_corrected} = 4.01 + 0.001 \times (T - 25)$$

$$\text{pH}_6\text{\_corrected} = 6.86 + 0.003 \times (T - 25)$$

$$\text{pH}_9\text{\_corrected} = 9.18 + 0.022 \times (T - 25)$$ [Used only if three-point calibration is employed]

Brief explanation: at $T = 25°C$, corrected values equal nominal buffer values.

3. pH Segment 1 (CI_4 to CI_6): For normal hydroponic operation, pH is interpolated between calibration points CI_4 and CI_6.

$$\text{pH} = \text{pH}_4\text{\_corrected} + (\text{pH}_6\text{\_corrected} - \text{pH}_4\text{\_corrected}) \times (\text{CI} - \text{CI}_4) / (\text{CI}_6 - \text{CI}_4)$$

Brief explanation: if $\text{CI} = \text{CI}_4$, output is $\text{pH}_4\text{\_corrected}$; if $\text{CI} = \text{CI}_6$, output is $\text{pH}_6\text{\_corrected}$

4. pH Segment 2 (CI_6 to CI_9, optional): If three-point calibration is enabled and **`CI`** is above CI_6, interpolation switches to the upper segment.

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pH = pH_6_corrected + (pH_9_corrected − pH_6_corrected) × (CI − CI_6) / (CI_9 − CI_6)
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$$pH = pH_{6}\_corrected + (pH_{9}\_corrected - pH_{6}\_corrected) \times (CI - CI_{6}) / (CI_{9} - CI_{6})$$

Brief explanation: this preserves continuity at CI_6 and improves estimation above the normal operating band.

5. Implementation Guard Conditions: Firmware checks denominator terms before division and handles invalid states.

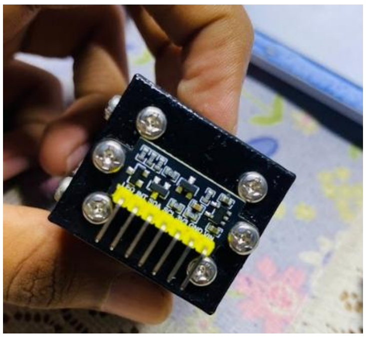
If $(CI_6 - CI_4) = 0$ or $(CI_9 - CI_6) = 0$, measurement is flagged invalid and recalibration is required.

6. Dye Volume and Water Level: The indicator is dosed at 7% v/v relative to the practical fill volume of the chamber. For a practical fill of 10.25 mL (near full capacity based on chamber dimensions), this gives 0.718 mL of working solution per test. In the sealed rectangular chamber (42×20×12.2mm), 10.25 mL corresponds to an approximate water level of 12.20mm from the chamber bottom.

V_dye = 0.07 × V_fill (0.718 mL per test @ 10.25 mL fill)

Brief explanation: 7% dosing keeps indicator concentration consistent across repeated measurements.

View of Final Product



The mechanical design of the optical pH sensor consists of a 6-part flow cell assembly: four PETG 3D-printed parts and two separate acrylic optical windows. The horizontal split architecture allows tube insertion without side-loading fragile components, provides a flat gasket sealing surface, and enables the chamber to be opened and cleaned in under five minutes.

5. Project Progress

Completed Tasks:

- Hardware design finalised: 4-part PETG flow cell CAD verified against drawings and STL files. All dimensions confirmed (body halves 55×16×28mm, LED mount 3mm, sensor mount 3.5mm).
- Optical system assembled: Dual LED system (6500K cool white + 3000K warm white) installed in left cap. TCS34725 MD0670 sensor mounted in sensor mount recess with foam tape.
- Electronics wiring complete: ESP32, TCS34725, and dual LED circuit connected and tested on breadboard.
- Pump timing analysed: NKP-DC-S06B pump flow rate (37 mL/min) evaluated. Dye pump assigned 35% PWM (ledcWrite = 89) to achieve reliable 0.718mL dosing over approximately 3.31s (>1 rotor revolution).
- Chemical preparation documented: BTB + Methyl Red indicator system documented with corrected dosing (0.718mL per test, approximately 139 tests per 100mL batch, approximately Rs. 0.23 per test).
- Comprehensive project documentation completed: Assembly guide, pump calibration guide, chemical preparation guide, pH color reading guide, BOM, and progress report all produced.

In Progress:

- **Leak sealing and chamber commissioning:** Water testing of the assembled flow cell identified a leak at the split face join. A 1mm silicone gasket is required between the top and bottom halves to provide a watertight seal. Silicone sealant is also needed at the three port fitting interfaces. This is the current critical path item before dye calibration can proceed.
- **Pump procurement and fluidic system installation:** 3× NKP-DC-S06B 12V peristaltic pumps, 4mm ID silicone tubing (2m), and 6× barbed fittings are pending procurement. Once installed, the pump timing calibration procedure (Section 4 of Pump Calibration Guide) will be executed for each pump individually.
- **pH buffer solutions preparation**: Three pH buffer sachets (pH 4.01, pH 6.86, and pH 9.18 at 25°C) dissolved in 250mL distilled water each and verified for color response.

Next Steps / Pending Tasks:

- Complete chamber sealing using 1mm silicone sheet gasket on the split face and aquarium silicone sealant at all port fittings. Perform a water-only leak test before introducing any indicator solution.
- Install and calibrate all three peristaltic pumps following the Pump Calibration Guide: measure actual flow rate for each pump, calculate timing constants (WATER_FILL_MS, DYE_PUMP_RUN_MS, DRAIN_RUN_MS, WATER_RINSE_MS), and verify dye dose accuracy to ±6% using a 0.01g precision scale.
- Prepare the BTB + Methyl Red working solution in collaboration with the University Chemistry Department. Validate the prepared solution against all three pH buffer solutions before sensor calibration.

- Execute full two-point (and optionally three-point) pH calibration: record baseline, pH 4.01 calibration point, pH 6.86 calibration point, and optionally pH 9.18. Verify calibration by re-testing pH 6.86 buffer and confirming reported pH reads within ±0.05.
- Perform 100-cycle reliability testing over multiple days to assess measurement consistency, dye degradation rate, and chamber cleanliness after drain and rinse cycles.
- Assess long-term stability over a 30-day continuous operation period with hydroponic nutrient solution to validate the ±0.025 pH accuracy target under realistic field conditions.

6. Components Used and Technologies

Hardware Components

- **ESP32 DevKit V1 (ESP-WROOM-32, 30-pin):** The central microcontroller used to run the FSM control logic, manage I2C communication with the TCS34725, control the three pump MOSFETs, read the DS18B20 temperature sensor via 1-Wire, store calibration values in EEPROM, and output pH readings via Serial (115200 baud).
- **TCS34725 RGB Color Sensor (MD0670 Rectangle):** The primary optical sensor. Communicates via I2C (address 0x29) at GPIO 21/22. Provides 16-bit resolution per channel (R, G, B, Clear) with a built-in IR blocking filter. The two onboard white LEDs are disabled by pulling the LED pin LOW at GPIO 4 (see Pin Assignment Table below for complete GPIO mapping), ensuring only the external dual LED system illuminates the sample.
- **Dual White LEDs (6500K Cool White + 3000K Warm White, 5mm):** Used as the broadband light source for transmissive colorimetry. Both LEDs are wired in parallel with individual 100Ω current-limiting resistors to GPIO 2. The dual-temperature configuration achieves ~CRI 85 equivalent, improving sensitivity for Methyl Red (red–yellow transition) and Bromothymol Blue (yellow–blue transition) simultaneously.
- **NKP-DC-S06B 12V Mini Peristaltic Pumps (×3):** Used for automated fluid handling. Pump 1 (GPIO 16) fills the chamber with sample water at 100% speed. Pump 2 (GPIO 12) injects the BTB+MR indicator at 35% PWM via ledcWrite to achieve a reliable 0.718 mL dose. Pump 3 (GPIO 18) drains the used sample. Pump 1 also handles the post-measurement rinse cycle, eliminating the need for a fourth pump.
- **IRFZ44N N-Channel MOSFET (×3):** Switches the 12V DC pump motors from the 3.3V GPIO logic of the ESP32. Suitable for gate-drive applications when driven at 3.3V logic levels; operation has been verified on breadboard with the specified pump load. Each MOSFET gate is connected via a 10kΩ pull-down resistor to ensure the gate stays LOW when the GPIO is floating. A 1N4007 flyback diode protects each MOSFET from back-EMF generated when the pump motor is switched off.

Pin Assignment Table

Signal / Peripheral	GPIO Pin	Protocol / Mode	Notes
Pump 1 (Fill)	GPIO 16	Digital Output (PWM)	100% duty cycle for sample fill
Pump 2 (Dye Inject)	GPIO 12	Digital Output (LEDC PWM)	35% PWM (~13 mL/min) for reliable 0.718 mL dosing
Pump 3 (Drain)	GPIO 18	Digital Output (PWM)	100% duty cycle for waste removal
Dual White LEDs	GPIO 2	Digital Output	Parallel drive with 100Ω limiting resistors per LED
TCS34725 Data (SDA)	GPIO 21	I2C	Shared I2C bus at 400 kHz
TCS34725 Clock (SCL)	GPIO 22	I2C	Shared I2C bus at 400 kHz

Assumptions and Validity Limits

- **Operational pH Range:** 4.0–7.6 (dual-segment calibration using pH 4.01 and pH 6.86 buffer reference points). Extension to pH 9.18 is pending optional three-point calibration and validation.
- **Temperature Compensation Valid Range:** 15–40°C (DS18B20 ±0.5°C accuracy). pH reference values are corrected for temperature drift within this range before interpolation.
- **Indicator Solution Batch Validity:** 6–9 months at room temperature when stored in darkness. Calibration must be repeated when indicator batch is replaced.
- **Recalibration Required:** After firmware update, sensor replacement, optical window damage or fouling, or when drift exceeds ±0.1 pH over three consecutive measurements against the same buffer.
- **Measurement Accuracy Status:** ±0.025–0.03 pH is a design target pending completion of 100-cycle reliability testing and 30-day field validation in hydroponic nutrient solution. Interim accuracy statement: ±0.05 pH (2σ, estimated budget) until full validation is complete.
- **Measurement Cycle Time:** Target <70 seconds per measurement. Nominal cycle time (10.25 mL operation): FILL (16.6 s) + DOSE (3.3 s) + SETTLE (2 s) + MEASURE (10 s, 20-sample averaging) + DRAIN (16.6 s) + RINSE (16.6 s) = ~65 seconds (subject to pump calibration confirmation).
- **M3 Brass Heat-Set Inserts and Stainless Steel Bolts:** 12× M3×3×4.2mm brass knurled heat-set inserts provide strong, reusable threaded connections in the PETG body. Assembly uses 4× M3×10mm bolts (body split join) and 8× M3×12mm bolts (cap joins). 2× M2 heat-set inserts and 2× M2×6mm bolts secure the TCS34725 PCB in its recess.
- **1mm Silicone Sheet (100×100mm) and Aquarium Silicone Sealant:** The silicone sheet is cut to 54×28mm and placed in the gasket groove on the body split face to create a watertight seal. Aquarium-grade silicone sealant is applied to all three port fitting interfaces (water inlet, dye port, drain port).

- **Clear PMMA Acrylic Rod (Ø12mm, 250mm length):** Cut into two 6mm pieces to form the optical windows. The windows are seated in split pockets (3mm deep in each half) and define an effective 30mm optical path in the sealed rectangular chamber. Surfaces are polished progressively using 400, 800, and 1000 grit sandpaper to achieve optical clarity.

Technologies

- **C/C++ (Arduino IDE / ESP32 Arduino Core):** For ESP32 firmware development. The firmware implements a deterministic Finite State Machine (IDLE → FILL → DOSE → SETTLE → MEASURE → DRAIN → RINSE → ERROR) with timeout protection and sensor read validation.
- **I2C Communication Protocol:** Standard two-wire serial interface (SDA at GPIO 21, SCL at GPIO 22) connects the ESP32 to the TCS34725 sensor. The Wire.h library handles all I2C transactions at 400kHz Fast Mode, with the Adafruit_TCS34725 library providing high-level sensor control.
- **ESP32 LEDC PWM (ledcSetup / ledcWrite):** Used to reduce Pump 2 (dye injection) speed to 35% of full speed (~13 mL/min from 37 mL/min rated flow rate). For a 0.718 mL dye target, this extends the dye dose run time from approximately 1.16 s at full speed to approximately 3.31 s, improving dosing repeatability and achieving ±5% dosing accuracy. Pump 1 (fill) and Pump 3 (drain) operate at 100% duty cycle without PWM modulation.
- **EEPROM (Persistent Calibration Storage):** The ESP32 EEPROM library stores all calibration values — baseline R/G/B, CI at each pH calibration point, and temperature-corrected reference pH values — in non-volatile memory. Calibration survives power cycles and only needs to be repeated when the indicator solution batch is replaced.
- **Piecewise Linear pH Calibration Model:** Two calibration segments are implemented in firmware: Segment 1 covers pH 4.01 to 6.86 using CI_4 and CI_6 reference points (mandatory for normal operation); Segment 2 covers pH 6.86 to 9.18 using CI_6 and CI_9 reference points (optional, used only if three-point calibration has been performed). This two-segment approach allows the sensor to handle the non-linear combined indicator response across the operational pH 4.0–7.6 hydroponic range and beyond.

Work Distribution

- Literature review and optical pH sensing approach research: Individual — 1 week
- CAD modelling of flow cell (4 parts, drawings, STL export): Individual (with CAD tool support) — 1 week
- Hardware procurement, assembly, and wiring: Individual — 2 weeks
- Chemical documentation and indicator formulation research: Individual — 1 week
- Documentation (assembly guide, calibration guide, chemical guide, BOM): Individual — 4 days
- Pump installation and calibration: Individual — pending (Week 7)
- Chemical preparation (BTB + MR stocks): University Chemistry Department collaboration — pending (Week 7)
- Full system integration testing and 100-cycle reliability validation: Individual — pending (Weeks 8–13)